hand, and Corsica, the Balearic Islands, Sardinia and the Tuscan archipelago on the other. Traffic intensity increases significantly during the summer months for passenger transport between the Mediterranean islands and the mainlan d, as well as with additional connections to North Africa and Barcelona and with cruise activity (see figures 5 and 6).

1.7 More than two thirds of the vessels using the study area (68% in winter and 71% in summer) fly a European flag, representing more than 70% of the cumulative distances travelled, whatever the season.

Figure 5: Representation of the maritime traffic during the winter period (2019, AIS source).

Figure 6: Representation of the maritime traffic during the summer period (2018, AIS source).

Harmful substances carried

1.8 The rules on the transport of harmful substances are derived from the International Convention for the Prevention of Pollution from Ships, known as the MARPOL Convention.

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SOURCE: MEPC.380(80).pdf
Harmful substances carried

1.8 The rules on the transport of harmful substances are derived from the International Convention for the Prevention of Pollution from Ships, known as the MARPOL Convention. These rules are contained in different international codes, depending on the nature and mode of transport of these substances. The Mediterranean is an important transport route, but also a major oil loading and unloading centre. It is also a major route for tankers.

1.9 In 2006, about 18% of the world's crude oil shipping, representing 4,224 voyages and 421 million tonnes, took place in the Mediterranean (MIU, 2008). Of the 10 main ports of discharge identified in 2006, 4 are located in the study area: Fos and Port-de-bouc (Marseille region), Genoa and Savona (Italy).

1.10 In 2006, liquefied natural gas (LNG) and liquefied petroleum gas (LPG) loadings amounted to 31 and 19 million tonnes respectively and unloading to 25 and 20 million tonnes

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SOURCE: MEPC.380(80).pdf
region), Genoa and Savona (Italy).

1.10 In 2006, liquefied natural gas (LNG) and liquefied petroleum gas (LPG) loadings amounted to 31 and 19 million tonnes respectively and unloading to 25 and 20 million tonnes for the whole Mediterranean (MIU, 2008).

1.11 Transported chemicals include organic compounds, animal oils and fats, inorganic compounds and other miscellaneous products. The transport of chemicals in liquid and gaseous form represents a relatively small share of international maritime trade (abo ut 2%) but remains a very dynamic and important sector in terms of value of goods; however, their accidental release would be harmful to the marine environment.

2 Natural factors

Hydrographical

2.1 Some areas are known to present risks to navigation, due to the presence of the narrowness of the passage or sectors with numerous islands and islets. This is particularly the case of the Strait of Bonifacio, which is 15 to 20 km wide and 100 m deep at its deepest point between southern Corsica and northern Sardinia. At its eastern mouth, it also contains the islands of the archipelagos of La Maddalena, Lavezzi and the island of Cavallo. This passage

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SOURCE: MEPC.380(80).pdf
between southern Corsica and northern Sardinia. At its eastern mouth, it also contains the islands of the archipelagos of La Maddalena, Lavezzi and the island of Cavallo. This passage is considered dangerous due to the presence of numerous rocks and strong currents that can increase the risk of grounding and other accidents. These characteristics prompted the establishment of the Bonifacio Strait PSSA.

2.2 The small pass of the islands of Hyères is also a potentially dangerous area for large vessels. Located between the Giens peninsula and the island of Porquerolles, its narrowest part extends over less than one mile, with depths of less than 20 m. The t raffic of passenger HSC is very important in the summer season. Cruise ships and ro-ro passenger ships also use it, generally in an east-west direction in heavy westerly weather (GIS3M, 2010).

Meteorological

2.3 The Mediterranean climate is characterized by hot, dry summers under the influence of the Azores anticyclone, and mild, relatively rainy winters. Local winds are variable, in both direction and strength, and become stronger in winter with gusts that ca n exceed 100 km/h.

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SOURCE: MEPC.380(80).pdf
of the Azores anticyclone, and mild, relatively rainy winters. Local winds are variable, in both direction and strength, and become stronger in winter with gusts that ca n exceed 100 km/h. North and north-west winds (Tramontane and Mistral) create the most violent storms.

Oceanographic

2.4 In the Mediterranean, the influence of the tides is weak; the tidal range does not exceed 40 cm on average near the coast. Tidal currents are weak and negligible compared to wind-induced currents. Generally, they are not felt near the coast in wide open areas, but they can be rapid in some narrow passages or shallow areas. The average sea waves and swell are generally weak, due to the small size of the Mediterranean basin where swells are infrequent and not very developed. The strongest states of the se a, in terms of height, are generated by north to north-west winds.

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3 Other Information: Impacts of shipping traffic on the area

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3 Other Information: Impacts of shipping traffic on the area

Collisions between ships and large cetaceans

3.1 The impact of ship collisions is now internationally recognized as an important threat to cetaceans, especially as shipping traffic, vessel size and speed continue to increase. Collisions involve a wide variety of vessels, with the risk of collision in creasing with vessel speed (as does the severity of injury to the animal), although there is currently insufficient data to adequately quantify this risk (Leaper, 2019).

3.2 The actual total number of collisions between large cetaceans and ships and the consequent impact at population level are difficult to be assessed. Accidents generally take place offshore and are rarely noticed by seafarers (this is particularly true w hen the vessels are large). Nevertheless, scientific work carried out over the last 15 years, sometimes in collaboration with shipping companies, has shown that two species are mainly concerned in

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SOURCE: MEPC.380(80).pdf
are large). Nevertheless, scientific work carried out over the last 15 years, sometimes in collaboration with shipping companies, has shown that two species are mainly concerned in the Mediterranean: the fin whale (Balaenoptera physalus) an d the sperm whale (Physeter macrocephalus). The latter spends long periods of rest floating at the surface, usually about 10

minutes, between deep dives: this behaviour makes it very vulnerable to ship strikes (UNEP/MAP-RAC/PSA, 2016).

3.3 Analysis of records of collisions between ships and the Mediterranean fin whale population over the period 1971-2001 showed that more than 80% of fatal ship strikes occurred in the North -West Mediterranean (Panigada et al., 2006). During the period 201 2-2018, the annual number of deadly collisions within the PSSA perimeter was up to 25.38 (Standard Deviation (SD) =5.97) fin whales per year. Based on recognized management rules, this value means that collisions alone prevent the restoration of the fin wh ale subpopulation within 100 years. Furthermore, there is almost a 10% chance that ship strike mortality triggers a subpopulation decline.

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means that collisions alone prevent the restoration of the fin wh ale subpopulation within 100 years. Furthermore, there is almost a 10% chance that ship strike mortality triggers a subpopulation decline.

3.4 Strandings data may complement the information on these accidents. A study carried out on strandings on the French coast since 1972 (Peltier et al., 2019) gave the following results:

.1 collisions are the main human cause of death for fin whales in the western Mediterranean (22.5% of stranding causes analysed on average; they are the cause of one in five strandings for all species combined);